

NEPAL COLLEGE OF INFORMATION TECHNOLOGY
ASSESSMENT FALL-2025

Level: Bachelor

Semester: Fall

Year: 2025

Program: BE SE_CE_VII

Full Marks: 100

Course Code: MGT 332

Pass Marks: 45

Course : Entrepreneurship and Professional Practice

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define entrepreneurship and explain its significance in the context of ICT-based ventures. 7
b) Describe the role and importance of entrepreneurship in economic development and technological innovation. 8
2. a) Differentiate between product innovation, process innovation, and business model innovation with examples. 7
b) Define Minimum Viable Product (MVP). Explain its role in reducing startup risk. 8
3. a) Explain the process of competitor analysis. How can startups gain competitive advantage? 7
b) Explain TAM, SAM, SOM, and GTM strategy with a suitable example. 8
4. a) Explain the Business Model Canvas and its key building blocks. 7
b) Discuss the importance of a business plan for startups and investors. 8
5. a) Explain the concept of whistleblowing and its ethical implications. 7
b) Discuss ethical challenges faced by technology startups and ways to address them. 8
6. a) Explain intellectual property rights (IPR) and their importance for startups. 7
b) Discuss the Electronic Transaction Act and its impact on digital business. 8

or

7. Write short notes on: (Any two)

- a) Innovation and Creativity
- b) Cash flow management
- c) Lean Canvas Model

NEPAL COLLEGE OF INFORMATION TECHNOLOGY
ASSESSMENT FALL 2025

Level: Bachelor
Program: BECE_(D/M) 7th
Course: Data Science and Analytics

Year: 2026
Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the concept of Knowledge Discovery from Data (KDD) and how it relates to the Data Science Pipeline. 7
- b) Differentiate between structured and unstructured data and Distinguish between numeric and categorical variables. 4+4
2. a) Explain Pearson correlation and how it measures relationship between two numeric variables show it by considering the example. 7
- b) Illustrate the empirical distribution of numeric and categorical data using the following: 8
 1. For numeric: Draw and interpret a histogram. Explain the significance of normal and power-law distributions.
 2. For categorical: Use a bar plot and explain Zipf's law with an example.
3. a) A real estate analyst collected data from 3 houses in a neighborhood to predict house prices based on two features: square footage (in 100s of square feet) and number of bedrooms. 7

Square footage(X1)	Number of bed rooms(X2)	House price (in \$1000s) (Y)
15	3	240
18	4	290
20	3	310

Find the regression equation: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

- b) A data analyst builds a multivariate linear regression model to predict house prices based on features such as square footage, number of bedrooms, and distance to the city center. 3+5
 1. Explain the significance of R² and Adjusted R² in evaluating the performance of the model. Why might Adjusted R² be a better metric when dealing with multiple predictors?
 2. While analyzing the model, the analyst suspects that some predictors (e.g., square footage and number of rooms) are highly correlated. (Explain in term of multicollinearity and how can it affect the model)
 3. Describe how Variance Inflation Factor (VIF) is used to detect multicollinearity and interpret what high VIF values imply.
- a) Derive the Nadaraya-Watson kernel regression estimator for the conditional mean $E[Y|X=x]$. Clearly show how the formula is obtained using kernel weighting. The performance of the Nadaraya-Watson estimator heavily depends on the choice of the bandwidth parameter. Explain the role of the bandwidth in kernel regression and discuss strategies for selecting an optimal bandwidth 7

(e.g., cross-validation, rule-of-thumb methods).

8

- b) You are given the following two-dimensional data set consisting of 6 observations:

(2,1),(3,5),(4,3),(5,6),(6,7),(7,8)

Using the Principal Component Analysis (PCA) algorithm, compute the first principal component of the dataset

5. a) You are given the following dataset of customers, where the target variable is whether the customer purchased a product (Yes or No): 7

Age	Income	Purchased
young	high	no
young	medium	yes
middle	high	yes
senior	Low	no
senior	medium	yes
middle	Low	yes

Using the CART (Classification and Regression Tree) algorithm, construct a binary classification tree (up to two levels deep) for predicting whether a customer will purchase the product.

- b) Apply the K-Means algorithm with $K=2$ on the following dataset: (160,55), (175,65), (172,60), (180,70), (178,68), (165,58) Use the first two data points as the initial centroids: and (160,55) and (175,65) Perform the clustering for two iterations, and at each step: 8

- Assign points to the nearest centroid
- Recompute the centroids
- Show the cluster membership after each iteration

6. a) What is time series analysis? Explain four components of time series: trend, seasonal variation, cyclical variation and irregular variation. Describe autocorrelation and stationarity with respect to time series analysis. 3+4

- b) Explain the differences between the ARMA(Auto-Regressive Moving Average) and ARIMA(Auto-Regressive Integrated Moving Average) models. Define the components of each model: Auto-Regressive (AR) Moving Average(MA) Integration (I). 8

7. Write short notes on: (Any two) 2×5

- permutation tests, partial and partial dependence plots
- Causation vs. correlation
- Statistical Significance: p-value
- logistic regression

NEPAL COLLEGE OF INFORMATION TECHNOLOGY
Assessment Fall 2025

Level: Bachelor
Programme: BE IT (M/D) VII
Course: : Engineering Economics (N)

Year : 2025
Full Marks: 100
Pass Marks: 45
Time : 3 Hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- 1) A) Describe the law of demand, law of supply. Production and utility theory? 7

B) How do you measure following cost concept in your engineering project; direct cost, indirect cost, average cost, variable cost, marginal cost, sunk cost, opportunity cost? 8

- 2) A) Ram Chandra is planning to retire in 20 years. He wishes to deposit regular money every 3 months until he retires so that he will receive annual payments of Rs 4,00,000 after the 1st year of his retirement for the next 10 years. How much must he deposit if the interest is 7%, compounded monthly? 7

B) Calculate both types of BCR using AW and FW formulation for the following projects. 8

Initial Investment	550000
Annual Revenue	15000
Annual Cost	20000
Useful life	8 yrs
Salvage Value	60000
MARR	12%

- 3) A) Using co-terminated assumption recommend the best project taking study period as 5 years. 8

Project	A	B
Initial Investment (RS)	350000	500000
Annual Revenues (Rs)	130000	175000
Annual cost (Rs)	15000	25000
Salvage value	35000	50000
Useful life	5 years	8 years
MARR	10%	10%

OR

Recommend the best project from the following two project.
Assume repeatability.

Project	A	B
Initial Investment	400000	700000
Annual revenues	175000	250000
Annual cost	25000	35000
Salvage Value	40000	70000
Useful life	6 yrs	8 yrs
MARR	12%	12%

B) What do you means by economic and financial analysis?
Determine IRR of the project with following information,

7

Initial investment= Rs 10,000

Project life= 7 years

Annual revenue= Rs 5550

Annual expenses= 3200

Salvage value= 2100

MARR= 7%

- 4) A) Perform sensitivity analysis of the following project over a range of $\pm 30\%$ in a) Initial Investment b) Net annual income and c) Useful life using PW formulation. MARR=10%. Draw sensitivity diagram also.

8

Initial Investment	25000
Annual Revenue	5500
Annual Cost	1500
Useful life	8 years
Salvage value	2500

B) From the following information, find the economic service life of an assets. 7

Initial cost= 55000

Annual Operating cost = 12000 for the 1st year and increases by 20% each year

Salvage value = Declining each year by 25% from the previous year's salvage value

Useful life 6 years

MARR= 15%

OR

Explain the reasons for replacement of assets. The annual equivalent cost(AEC) of defender and challenger are given in the table below. What is the best replacement strategy? Use MARR: 10% per year and the planning horizon of the project is 7 years.

EOY	1	2	3	4	5	6
(AEC)D RS	5300	5250	5400	5750	6200	6550
(AEC)C RS	7700	6150	5700	5600	5675	5800

5) A) Evaluate whether the project is feasible for investment or not using payback methods? 7

Find both types of payback period.

Initial investment= 450000

Annual benefit= 80000

Annual cost= 25000

Salvage value= 110000

Useful life= 10 years

Interest rate= 10% per year

B) Explain cost of capital. Calculate WACC of following details: 8

Interest on debt = 12%

Dividend on equity = 15%

Debt = 70%

Equity = 30%

Tax rate = 20%

- 6) A) What are the causes of depreciation? If a machine costing Rs. 250000 is purchased by expecting salvage value of Rs 60000 at the end of 6th year. Calculate depreciation amount for each year by 8

i) SOYD

ii) Declining balance

B) Define inflation and its causes. Explain actual dollar and constant dollar. 7

7) Write short notes on: (Any two) 2*5

a) Depreciation and depletion

b) Simple, compound and nominal interest rate

c) Direct and indirect tax

Nepal College of Information Technology
Assessment Fall 2025

Level: Bachelor
Programme: Computer Engineering
Course: Network and Cyber Security

Year : 2025
Full Marks : 100
Time : 3 hrs

Candidates are required to give answers in their own words as far as practicable. Attempt ALL questions. Figures in the right margin of the questions indicate their full marks.

- 1 (a) Describe with an example each of an active and a passive attack. 7
{Chapter1}
- (b) Encrypt the plaintext "cyber security is fun" using Caesar cipher 8
using the key equal to the last two digits of your class roll
number mod 26. {Chapter2}
- 2 (a) Explain the operation of S-Box of DES with an example. 7
{Chapter2}
- (b) Elucidate the key generation schedule of AES. {Chapter2} 8
- 3 (a) What do you mean by Prime Factorization problem? Explain in 7
context of RSA algorithm. {Chapter3}
- (b) Show that Diffie-Hellman establishes identical keys at both the 8
communicating principals. {Chapter3}
- 4 (a) Geometrically elucidate the Point Adding and Point Doubling 7
operations in ECC. {Chapter3}
- (b) Write down the I/O properties and security properties of a hash. 8
{Chapter4}
- 5 (a) How is a digital signature verified according to the Digital 7
Signature Standard? {Chapter4}
- (b) How does a man in the middle of sender and recipient take 8
control of the entire communication? {Chapter4}

- 6 (a) Explain the stages of SSL handshaking protocol with necessary diagrams. {Chapter5} 7
- (b) How does Radix-64 convert arbitrary binary data into canonical printable ASCII format? {Chapter5} 8

OR

- (c) Differentiate with necessary diagrams the IP packet encapsulation in tunneling and transport mode. 8
7. Write short notes on any TWO only {Chapter6} 5x3
- (a) Broken access control
- (b) Authentication and Authorization
- (c) Error handling and Exception management

NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Assessment Fall 2025

Level: Bachelor
Programme: BE Computer
Course: Cloud Computing and Virtualization

Semesters: Fall
Year: 2025
Full Marks: 100
Pass Marks: 45
Time: 2 Hrs

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define cloud computing according to the NIST definition. 3+4
Discuss the evolution of cloud computing and its impact on modern IT infrastructure.
- b) Compare and contrast public, private, hybrid, and community cloud deployment models with examples. 8
2. a) Explain virtualization technologies and their role in enabling cloud computing. 7
- b) Describe the architecture of a Virtual Private Cloud (VPC) including subnets, gateways, and DNS services 8
3. a) Discuss redundancy and high availability in cloud storage systems. 3+4
- b) Explain scaling strategies in cloud computing with examples of auto-scaling and load balancing. 8
4. a) Explain Identity and access management (IAM). 7
- b) Explain Service-level agreement (SLA) with its types. 8
5. a) What are the major cloud security threats? Suggest mechanisms to mitigate them. 7
- b) List three types of cloud storage models and give one use case for each. 8
6. a) Explain the Jericho Cloud Cube model. 7
- b) Discuss disaster management strategies in cloud computing environments. 8
7. Write short notes on: (Any two) 2x5
- b) Containers
- a) Hardware Hypervisor
- b) Billing management systems